Vectors — Worksheet

Chapters 21 & 22: Solved Examples (questions only) and Exercises

CHAPTER 21 — VECTORS

Illustrative Solved Examples (Questions)

Example 1. If $\vec{a}$, $\vec{b}$ are the vectors determined by two adjacent sides of a regular hexagon, what are the vectors determined by the other sides taken in order. [ISC]

Example 2. $ABCDEF$ is a regular hexagon. $\overrightarrow{AB}$ is $\vec{a}$ and $\overrightarrow{BC}$ is $\vec{b}$; express $\overrightarrow{AC}$, $\overrightarrow{AD}$, and $\overrightarrow{AF}$ in terms of $\vec{a}$ and $\vec{b}$. [ISC]

Example 3. $PQRS$ is a quadrilateral. Find the resultant of the vectors $\overrightarrow{QP}, \overrightarrow{QR}, \overrightarrow{RS}$ and $\overrightarrow{SP}$ in term of $\overrightarrow{QP}$.

Example 4. If $|\overrightarrow{a}| = 5$, interpret (i) $\mathbf{6}\overrightarrow{a}$ (ii) $-7\overrightarrow{a}$.

Example 5. If the sum of two unit vectors is a unit vector, show that the magnitude of their difference is $\sqrt{3}$. [ISC]

Example 7. In a parallelogram $ABCD, AE$ is one-half of AB. Using vectors find $AF : AC$ (Fig. 21.18). [ISC]

Example 8. Let $\boldsymbol{O}$ be the origin and let $\boldsymbol{P}(-4, 3)$ be a point in the $\boldsymbol{xy}$-plane. Express $\overrightarrow{OP}$ in terms of vectors $\hat{i}$ and $\hat{j}$. Also, find $|\overrightarrow{OP}|$.

Example 9. $\vec{a}, \vec{b}$ are position vectors of the points $(1, -1), (-2, m)$, find the value of m for which $\overrightarrow{a}$ and $\overrightarrow{b}$ are collinear.

Example 10. If the position vector $\overrightarrow{a}$ of the point $(5, \beta)$ is such that $|\overrightarrow{a}| = 13$, find the value of $\boldsymbol{\beta}$.

Example 11. (i) Write the value of $|(6, 3) + (6, 2)|$.
(ii) If $\boldsymbol{P} = (2, 3)$ and $\boldsymbol{Q} = (5, 7)$, express the vector $\boldsymbol{PQ}$ in terms of the unit vectors $\hat{i}$ and $\hat{j}$. Also find $|\overrightarrow{PQ}|$.
(iii) If $\boldsymbol{A} = (0, 1), \boldsymbol{B} = (1, 0), \boldsymbol{C} = (1, 2), \boldsymbol{D} = (2, 1)$, prove that $\boldsymbol{AB} = \boldsymbol{CD}$.

Example 12. Find the coordinates of the tip of the position vector which is equivalent to $\overrightarrow{AB}$, where
(i) $A = (3, 1), B = (5, 0)$,
(ii) $A = (-1, 3), B = (5, 0)$.

Example 13. $\overrightarrow{a}$ is a position vector whose tip is (1,-3). Determine the coordinates of the point $\boldsymbol{B}$ such that $\overrightarrow{\boldsymbol{AB}} = \overrightarrow{a}$, if $\boldsymbol{A}$ has coordinates (-1, 5).

Example 14. Using vectors, show that the points $\boldsymbol{A}(-2, 1), \boldsymbol{B}(-5, -1)$ and $\boldsymbol{C}(1, 3)$ are collinear.

Example 15. Find a unit vector parallel to the vector $-3\hat{i} + \hat{j}$.

Example 16. $ABCD$ is a parallelogram and A, B, C have respectively the coordinates (2, 3), (1, 4) and (0, -2). Find the coordinates of D.

Example 17. Find the magnitude of the vector $\overrightarrow{\boldsymbol{a}} = 2\hat{\boldsymbol{i}} - 3\hat{\boldsymbol{j}} + \hat{\boldsymbol{k}}$.

Example 22. If the vertices of a triangle are (2, -1, 1), (1, -3, -5) and (3, -4, -4), prove by vector method that it is a right-angled triangle.

Example 23. If the position vectors of two points $\boldsymbol{A}$ and $\boldsymbol{B}$ are $\hat{\boldsymbol{i}} + 3\hat{\boldsymbol{j}} + 4\hat{\boldsymbol{k}}$ and $2\hat{\boldsymbol{i}} + 5\hat{\boldsymbol{j}} + 7\hat{\boldsymbol{k}}$ respectively, find the direction cosines of the vector $\overrightarrow{AB}$.

Example 24. If the position vectors of $\boldsymbol{P}, \boldsymbol{Q}, \boldsymbol{R}$ and $\boldsymbol{S}$ are $2\hat{\boldsymbol{i}}+4\hat{\boldsymbol{k}}, 5\hat{\boldsymbol{i}}+3\sqrt{3}\hat{\boldsymbol{j}}+4\hat{\boldsymbol{k}}, -2\sqrt{3}\hat{\boldsymbol{j}}+\hat{\boldsymbol{k}}$ and $2\hat{i}+\hat{k}$, prove that RS is parallel to PQ and is $\frac{2}{3}$ of PQ.

Example 25. The position vectors of two points $\boldsymbol{P}$ and $\boldsymbol{Q}$ are $\hat{\boldsymbol{i}} + \hat{\boldsymbol{j}} + \hat{\boldsymbol{k}}$ and $5\hat{\boldsymbol{i}} - 3\hat{\boldsymbol{j}} + \hat{\boldsymbol{k}}$ respectively. Show that PQ is parallel to $X - Y$ plane, and find its length also.

Example 26. The two adjacent sides of a parallelogram are $2\hat{\boldsymbol{i}} + 4\hat{\boldsymbol{j}} - 5\hat{\boldsymbol{k}}$ and $\hat{\boldsymbol{i}} + 2\hat{\boldsymbol{j}} + 3\hat{\boldsymbol{k}}$. Find the unit vectors along the diagonals of the parallelogram.

Example 27. Find a unit vector parallel to the sum of the vectors $2\hat{\boldsymbol{i}} + 4\hat{\boldsymbol{j}} - 5\hat{\boldsymbol{k}}$ and $\hat{i} + 2\hat{j} + 3\hat{k}$.

Example 28. Show that the points $\boldsymbol{A}(1, 2, 3), \boldsymbol{B}(3, 4, 7)$ and $\boldsymbol{C}(-3, -2, -5)$ are collinear.

EXERCISE 21 (a)

1. Draw a pair of directed segmems $\overrightarrow{AB}$ and $\overrightarrow{XY}$ wifich are parallel, in fine opposite sense, and equal in length. Find a directed segment which represents their sum. What can you say about by its length, (ii) its direction?

2. Simplify for triangles ABC and PQB
(i) $\overrightarrow{AB} + \overrightarrow{BA}$
(ii) $\overrightarrow{BC} + \overrightarrow{CA} + \overrightarrow{AB}$
(iii) $\overrightarrow{PQ} + \overrightarrow{PB} + \overrightarrow{QR}$

3. In Fig 21,21 the various line segments are taken to be representatives of weutors. Find from the figure a single representative of each of the following sums :
(i) $\overrightarrow{AE} + \overrightarrow{EC}$
(ii) $\overrightarrow{DB} + \overrightarrow{BE}$
(iii) $\overrightarrow{AD} + \overrightarrow{DB} + \overrightarrow{BC}$
(in) $\overrightarrow{CB} + \overrightarrow{BE} + \overrightarrow{EA} + \overrightarrow{AD}$

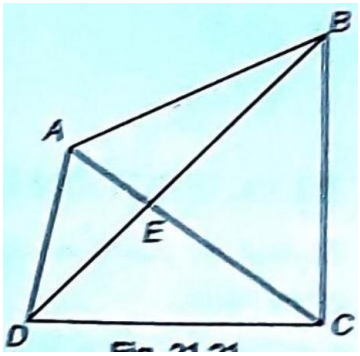


Fig. 21.21

4. In Fig 21.21, simplify:
(i) $\overrightarrow{DE} + (-\overrightarrow{BE})$
(ii) $\overrightarrow{AC} + (-\overrightarrow{BC})$
(iii) $\overrightarrow{CD} + \overrightarrow{BA}(-\overrightarrow{BD})$

5. In Fig 21.22, simplify :
(i) $\overrightarrow{AC} - \overrightarrow{AB}$
(ii) $\overrightarrow{BA} - \overrightarrow{BC}$
(iii) $\overrightarrow{BC} - \overrightarrow{BA}$
(iv) $\overrightarrow{CA} - \overrightarrow{CB}$
(v) $\overrightarrow{CB} - \overrightarrow{CA}$

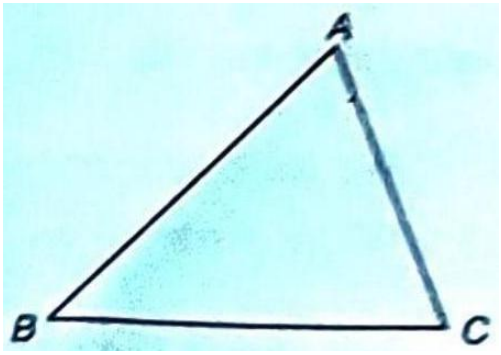


Fig. 21.22

6. In Fig. 21.23, EFGH is a parallelogram. Simplify :
(i) $\overrightarrow{EF} + \overrightarrow{EH}$
(ii) $\overrightarrow{EF} - \overrightarrow{EH}$
(iii) $\overrightarrow{EH} - \overrightarrow{EF}$
(iv) $\overrightarrow{FG} + \overrightarrow{FE}$
(v) $\overrightarrow{FG} - \overrightarrow{FE}$
(vi) $\overrightarrow{FE} - \overrightarrow{FG}$
(vii) $\overrightarrow{GE} - \overrightarrow{GH}$
(viii) $\overrightarrow{HG} - \overrightarrow{HE}$

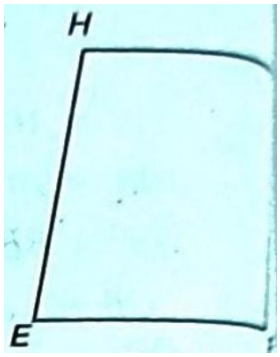


Fig. 21.23

EXERCISE 21 (b)

1. $\overrightarrow{OA}$ and $\overrightarrow{OB}$ are vectors $\vec{a}$ and $\vec{b}$ respectively and X and Y are points of trisection of AB. Find, in terms of $\vec{a}$ and $\vec{b}$,
(i) $\overrightarrow{OX}$ and
(ii) $\overrightarrow{OY}$.

2. $\overrightarrow{OA}$ and $\overrightarrow{OB}$ are vectors $\vec{a}$ and $\vec{b}$ respectively and P and Q are points $\frac{1}{4}$ and $\frac{3}{4}$ of the way along AB. Find, in terms of $\vec{a}$ and $\vec{b}$:
(i) $\overrightarrow{OB}$ and
(ii) $\overrightarrow{OQ}$.

3. $ABCD$ is a quadrilateral in which BC is parallel to AD and the ratio of the lengths $BC : AD$ is 4:7. Taking $\overrightarrow{AB}$ and $\overrightarrow{AD}$ as representatives of vectors $\vec{v}$ and $7\vec{u}$ respectively, find which vectors are represented by
(i) $\overrightarrow{BC}$
(ii) $\overrightarrow{AC}$
(iii) $\overrightarrow{BD}$
(iv) $\overrightarrow{DC}$;
(v) $\overrightarrow{AE}$ where E is on BD such that $BE = \frac{4}{11}BD$ in length;
(vi) $\overrightarrow{AF}$ where F is on AC such that $AF = \frac{7}{11}AC$.

4. In Fig. 21.30, BE is a median of triangle ABC and G divides BE in the ratio $2 : 1$.
(i) If $\overrightarrow{AB}$ represents $\vec{u}$ and $\overrightarrow{AC}$ represents $\vec{v}$, show that $\overrightarrow{EB}$ represents $\vec{u} - \frac{1}{2}\vec{v}$ and that $\overrightarrow{AG}$ represents $\frac{1}{3}(\vec{u} + \vec{v})$. [ISC]
(ii) If CF is a median, and H divides CF in the ratio $2 : 1$, show that $\overrightarrow{AH}$ represents $\frac{1}{3}(\vec{u} + \vec{v})$.

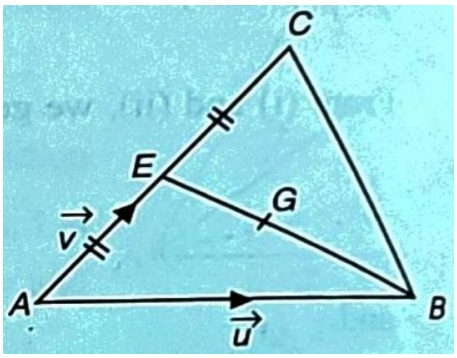


Fig. 21.30

(iii) If AD is a median and K divides AD in the ratio $2 : 1$, which vector does $\overrightarrow{AK}$ represent in terms of $\vec{u}$ and $\vec{v}$?
What can you conclude about G, H, K ? What can you conclude about the medians of a triangle ?

5. Four points A, B, C, D with position vectors $\vec{a}, \vec{b}, \vec{c}, \vec{d}$ respectively are such that $3\vec{a} - \vec{b} + 2\vec{c} - 4\vec{d} = \overrightarrow{0}$. Show that the four points are coplanar. Also, find the position vector of the point of intersection of lines AC and BD.

EXERCISE 21 (c)

13. If $\vec{a} = 7\hat{i} - 6\hat{j}$ and $\vec{b} = 3\hat{i} + 4\hat{j}$, find $\vec{a} + \vec{b}$ and determine a unit vector in the direction of the vector $\vec{a} + \vec{b}$.

14. Prove that the vectors $\vec{a} = -4\hat{i} - \hat{j}, \vec{b} = \hat{i} - 4\hat{j}$ and $\vec{c} = 3\hat{i} + 5\hat{j}$ form a right angled-triangle.

15. Show that the point $\vec{a} = -3\sqrt{3}\hat{i} - 3\hat{j}, \vec{b} = 6\hat{j}$, $\vec{c} = 3\sqrt{3}\hat{i} + 3\hat{j}$ form the sides of an isosceles triangle.

16. Show that the three points with position vectors $2\hat{i} + 3\hat{j}, 3\hat{i} + \frac{9}{4}\hat{j}, 5\hat{i} + \frac{3}{4}\hat{j}$ are collinear.

EXERCISE 21 (d)

1. If $\overrightarrow{OA} = 2\hat{i} - \hat{j} + 3\hat{k}$ and $\overrightarrow{OB} = 5\hat{i} + 2\hat{j} + 3\hat{k}$, then find the vectors $\overrightarrow{AB}$ and $\overrightarrow{BA}$.

2. If $\vec{a} = 4\hat{i} + \hat{j} + \hat{k}, \vec{b} = 2\hat{i} + \hat{j} - 7\hat{k}, \vec{c} = -3\hat{i} - 4\hat{j} + 2\hat{k}$ and $\vec{d} = \hat{i} + \hat{j} + \hat{k}$, then find the value of $3\vec{a} + 2\vec{b} - 4\vec{c} - \vec{d}$.

3. Find sum of the vectors $3\hat{i} + 7\hat{j} - 4\hat{k}$ and $\hat{i} - 5\hat{j} - 8\hat{k}$ and hence find the unit vector along the sum of these vectors.

4. Find by vector method the perimeter of the triangle whose vectors are (-1, -1, 9), (3, 1, 5) and (0, -5, 1).

5. If the position vectors of the angular points A, B, C and D of a quadrilateral be $\hat{i} + \hat{j} + \hat{k}$, $\hat{j} + \hat{k} - \hat{i}, \hat{k} + \hat{i} - \hat{j}$, and $\hat{i} + \hat{j} - \hat{k}$ respectively, then determine the lengths of its sides.

6. Prove by vector method that the triangle whose vertices are (2, 4, -1), (4, 5, 1) and (3, 6, -3) is an isosceles triangle.

7. The position vectors of two points A and B are $5\hat{i} + \hat{j} + \hat{k}$ and $3\hat{i} + 2\hat{j} + \hat{k}$ respectively. Find the direction cosines of $\overrightarrow{AB}$ and show that AB is parallel to $X - Y$ plane.

8. Given that $\vec{a} = (x + 4y)\hat{i} + (2x + y + 1)\hat{j}$ and $\vec{b} = (y - 2x + 2)\hat{i} + (2x - 3y - 1)\hat{j}$, find the value of x and y, if $3\vec{a} = 2\vec{b}$.

9. Prove that the vectors $2\hat{i} + 3\hat{j} - 6\hat{k}, 6\hat{i} - 2\hat{j} + 3\hat{k}$ and $3\hat{i} + 6\hat{j} - 2\hat{k}$ form the sides of an equilateral triangle.

10. Prove that the points $2\hat{i} - \hat{j} + \hat{k}, \hat{i} - 3\hat{j} - 5\hat{k}, 3\hat{i} - 4\hat{j} - 4\hat{k}$ are the vertices of a right-angled triangle.

11. If the point $\hat{i} - \hat{j} + \hat{k}, 2\hat{i} + 3\hat{j} + 4\hat{k}$ and $3\hat{i} + 7\hat{j} + p\hat{k}$ a re collinear, then find the value of p.

12. If the vertices of a triangle are the points $\hat{i} - \hat{j} + 2\hat{k}$, $2\hat{i} + 3\hat{j} + 4\hat{k}, 3\hat{i} + 3\hat{j} - 4\hat{k}$, what are the vectors determined by its sides? $\hat{i}, \hat{j}, \hat{k}$ are unit vectors parallel to the axes of coordinates.

13. Find the length and direction cosines of $\overrightarrow{PQ}$ where $\overrightarrow{OP} = -2\hat{i} + 3\hat{j} - 4\hat{k}$ and $\overrightarrow{OQ} = 2\hat{i} - \hat{j} + 3\hat{k}$.

14. The vertices of a triangle are $P(-2, 1, -3), Q(0, 4, 3)$, $R(2, 5, 4)$. Find the lengths and direction cosines of the vectors $\overrightarrow{PQ}$ and $\overrightarrow{PR}$.

15. A is the point $(1, 3, -2), B$ is $(3, -4, 1), C$ is $(-1, 0, 2)$. Find the lengths of the sides of the triangle ABC and the cosines of the angles, A, B, C.

16. Prove that the triangle with vertices $A(1, 0, 1), B(2, -1, 4)$ and $C(3, -4, -1)$, is right-angled.

17. The position vectors of A, B and C are $2\hat{i} + \hat{j} - \hat{k}$, $3\hat{i} - 2\hat{j} + \hat{k}$ and $\hat{i} + 4\hat{j} - 3\hat{k}$ respectively, show that A, B and C are collinear.

18. Show that the points (2, -1, 3), (3, -5, 1) and (-1, 11, 9) are collinear.

CHAPTER 22 — VECTORS (CONTINUED)

Section A: Scalar (Dot) Product — Solved Examples (Questions)

Example 1. Find $\vec{a} \cdot \vec{b}$ if (i) $\vec{a} = \hat{i} + 2\hat{j} + 3\hat{k}, \vec{b} = -2\hat{j} + 4\hat{k}$ (ISC)
(ii) $\vec{a} = (1, 1, 2)$ and $\vec{b} = (3, 2, -1)$.

Example 2. Find $(\vec{a} + 3\vec{b}) \cdot (2\vec{a} - \vec{b})$, if $\vec{a} = \hat{i} + \hat{j} + 2\hat{k}$ and $\vec{b} = 3\hat{i} + 2\hat{j} - \hat{k}$.

Example 3. Dot products of a vector with vectors $3\hat{i} - 5\hat{k}, 2\hat{i} + 7\hat{j}$ and $\hat{i} + \hat{j} + \hat{k}$ are respectively-1, 6 and 5. Find the vector.

Example 4. Two vectors $\vec{a}$ and $\vec{b}$ are such that $(\vec{a} + \vec{b})(\vec{a} - \vec{b}) = \mathbf{0}$. Show that $|\vec{a}| = |\vec{b}|$.

Example 5. If $\vec{a}$ and $\vec{b}$ be two non-zero vectors, show that $|\vec{a} + \vec{b}|^2 + |\vec{a} - \vec{b}|^2 = 2|\vec{a}|^2 + 2|\vec{b}|^2$. Interpret the result geometrically.

Example 6. If $\vec{a}$ is any vector, then $\vec{a} = (\vec{a} \cdot \hat{i})\hat{i} + (\vec{a} \cdot \hat{j})\hat{j} + (\vec{a} \cdot \hat{k})\hat{k}$

Example 7. If $\vec{a}$ and $\vec{b}$ are two vectors such that $|\vec{a}| = 4, |\vec{b}| = 3$ and $\vec{a} \cdot \vec{b} = 6$, find the angle between $\vec{a}$ and $\vec{b}$.

Example 8. Find the scalar product $\vec{a} \cdot \vec{b}$ of two vectors $\vec{a}$ and $\vec{b}$ and also the angle between them, if
(i)

$\vec{a} = 3\hat{i} - 2\hat{j} + \hat{k}$ and $\vec{b} = 2\hat{i} + 3\hat{j}$. (ii) $\vec{a} = 2\hat{i} + 2\hat{j} - \hat{k}$ and $\vec{b} = 6\hat{i} - 3\hat{j} +$

Example 9. Find the magnitude of $\angle BAC$, where A, B, C are the vertices of $\triangle ABC$ having position vectors as the points (2, 1, 3), (4, 1, 3) and (0, 2, 1) respectively.

Example 10. If $|\vec{a}| = 3, |\vec{b}| = 5, |\vec{c}| = 7$ and $\vec{a} + \vec{b} + \vec{c} = \mathbf{0}$, prove that the angle between $\vec{a}$ and $\vec{b}$ is $\pi/3$.

Example 11. If $\vec{a} + \vec{b} + \vec{c} = \mathbf{0}$, show that the angle $\boldsymbol{\theta}$ between the vectors $\vec{b}$ and $\vec{c}$ is given by $\cos\theta = \frac{\vec{a}^2 - \vec{b}^2 - \vec{c}^2}{2|\vec{b}||\vec{c}|}$.

Example 12. If $\vec{a}, \vec{b}$ are two unit vectors and θ is the angle between them, show that $\sin\frac{1}{2}\theta = \frac{1}{2}|\vec{a} - \vec{b}|$.
(ISC 2013, UPSEAT 2002)

Example 13. Find the angles which the vector $\vec{a} = \mathbf{3\hat{i} - 6\hat{j} + 2\hat{k}}$ makes with the coordinate axes.

Example 14. (i) Find the value of λ such that the vectors $2\hat{i} - \hat{j} - \hat{k}$ and $\lambda\hat{i} + \hat{j} + 5\hat{k}$ are perpendicular.
(ii) If $\vec{A} = \hat{i} + \mathbf{2}\hat{j} + \mathbf{3}\hat{k}$, $\vec{B} = \hat{i} + \mathbf{2}\hat{j} + \hat{k}$ and $\vec{C} = \mathbf{3}\hat{i} + \hat{j}$, then evaluate $\boldsymbol{t}$ if the vectors $(\vec{A} + \boldsymbol{r}\vec{B})$ and $\vec{C}$ are mutually perpendicular. (MNR)

Example 15. Find the value of p for which the vectors $\vec{a} = \mathbf{3\hat{i} + 2\hat{j} + 9\hat{k}}$ and $\vec{b} = \hat{i} + \boldsymbol{p}\hat{j} + \mathbf{3}\hat{k}$ are (i) perpendicular (ii) parallel.

Example 16. (i) Prove that the triangle whose vertices have the position vectors $2\hat{i}+4\hat{j}-\hat{k}$, $4\hat{i}+5\hat{j}+\hat{k}$, $3\hat{i}+6\hat{j}-3\hat{k}$ is an isosceles right-angled triangle.
(ii) Show that the vectors $\mathbf{2\hat{i} - \hat{j} + \hat{k}}, \hat{i} - \mathbf{3}\hat{j} - \mathbf{5}\hat{k}$ and $\mathbf{3\hat{i} - 4\hat{j} - 4\hat{k}}$ form the position vectors of the vertices of a right-angled triangle. Find the remaining angles of the triangle also. (ISC 2003, 1999)

Example 17. Show that $\frac{1}{7}(2\hat{i}+3\hat{j}+6\hat{k})$, $\frac{1}{7}(3\hat{i} \quad 6\hat{j}+2\hat{k})$ and $\frac{1}{7}(6\hat{i}+2\hat{j}-3\hat{k})$ are mutually perpendicular unit vectors. (MNR)

Example 18. Let $\vec{a}, \vec{b}, \vec{c}$ be three vectors of magnitudes 3, 4 and 5 respectively. If each one is perpendicular to the sum of the other two vectors, prove that $|\vec{a} + \vec{b} + \vec{c}| = 5\sqrt{2}$.

Example 19. The scalar product of the vector $\hat{i} + \hat{j} + \hat{k}$ with a unit vector along the sum of vectors $\mathbf{2\hat{i} + 4\hat{j} - 5\hat{k}}$ and $\boldsymbol{\lambda}\hat{i} + \mathbf{2}\hat{j} + \mathbf{3}\hat{k}$ is equal to one. Find the value of $\boldsymbol{\lambda}$.

Example 20. (i) If $(\vec{a} + \vec{b}) \cdot (\vec{a} - \vec{b}) = \mathbf{0}$, prove that $|\vec{a}| = |\vec{b}|$.
(ii) If $\vec{c}$ is normal to $\vec{a}$ and $\vec{b}$, show that $\vec{c}$ is normal to $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$.
(iii) If two vectors $\vec{a}$ and $\vec{b}$ be such that $|\vec{a} + \vec{b}| = |\vec{a} - \vec{b}|$, then prove that $\vec{a} \perp \vec{b}$.
Interpret the result geometrically. (Type ISC 2004, Karnataka CET 2007)

Example 21. If $\vec{a}, \vec{b}, \vec{c}$ are three non-coplanar vectors and $\vec{d} \cdot \vec{a} = \vec{d} \cdot \vec{b} = \vec{d} \cdot \vec{c} = \mathbf{0}$, then show that $\vec{d}$ is a null vector.

Example 22. If $\vec{a}$ and $\vec{b}$ are two perpendicular vectors, show that $(\vec{a} + \vec{b})^{\mathbf{2}} = (\vec{a} - \vec{b})^{\mathbf{2}}$.
(ISC)

Example 23. If $\vec{a} \cdot \vec{b} = \vec{a} \cdot \vec{c}$, show that either $\vec{a} = \mathbf{0}$ or $\vec{b} = \vec{c}$ or $\vec{a}$ is perpendicular to $(\vec{b} - \vec{c})$.

Example 24. Express the vector $\vec{a} = 5\hat{i} - \mathbf{2}\hat{j} + \mathbf{5}\hat{k}$ as the sum of two vectors such that one is parallel to the vector $\vec{b} = (3\hat{i} + \hat{k})$ and the other is perpendicular to $\vec{b}$.
(CBSE 2005)

Example 25. If $\hat{\boldsymbol{a}}, \hat{\boldsymbol{b}}, \hat{\boldsymbol{c}}$ are three mutually perpendicular unit vectors, show that $\hat{\boldsymbol{a}} + \hat{\boldsymbol{b}} + \hat{\boldsymbol{c}}$ is equally inclined to $\hat{\boldsymbol{a}}, \hat{\boldsymbol{b}}$ and $\hat{\boldsymbol{c}}$.
(ISC 2017)

Example 26. Find the projection of $\vec{a} = \hat{i} - \mathbf{2}\hat{j} + \hat{k}$ on the vector $\vec{b} = \mathbf{4\hat{i} - 4\hat{j} + 7\hat{k}}$.
(MNR), (Type ISC 2018)

Example 27. Find the vector projection of $\vec{b}$ on $\vec{a}$, when $\vec{a} = \hat{i} - \mathbf{2}\hat{j} + \hat{k}$, $\vec{b} = 7\hat{i} + \hat{j} - \mathbf{3}\hat{k}$.

Example 28. If $\vec{a} = \mathbf{4\hat{i} + 6\hat{j}}$ and $\vec{b} = \mathbf{3\hat{j} + 4\hat{k}}$, find the vector form of the component of $\vec{a}$ along $\vec{b}$. (MNR)

EXERCISE 22 (a)

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1. If $\vec{a} = \hat{i} + 2\hat{j} - \hat{k}, \vec{b} = 2\hat{i} + \hat{j} + \hat{k}$ and $\vec{c} = 5\hat{i} - 4\hat{j} + 3\hat{k}$, then find the value of
(i) $\vec{a}, \vec{b}$
(ii) $\vec{b}, \vec{c}$
(iii) $\vec{c} \cdot \vec{a}$
(iv) $|\vec{a}|$
(v) $|\vec{b}|$
(vi) $|\vec{c}|$
(vii) $\vec{a} \cdot (\vec{b} + \vec{c}), \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$ and show that they are equal.
(viii) $(2\vec{a} - \vec{b} + 2\vec{c}).\vec{c}$
(ix) $(\vec{a} - 2\vec{b}) \cdot (\vec{b} - 2\vec{c})$
(x) angle between $\vec{a}$ and $\vec{b}, \vec{b}$ and $\vec{c}$ and $\vec{c}$ and $\vec{a}$.

11. Find the projection of $\vec{b}+\vec{c}$ on $\vec{a}$, where $\vec{a} = 2\hat{i}-2\hat{j}+\hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} - 2\hat{k}$ and $\vec{c} = 2\hat{i} - \hat{j} + 4\hat{k}$. (CBSE 200)

12. Using vectors, find the projection of the line joining the points $(1, -1, 1)$ and $(-2, 3, 4)$ on the line joining the points (4, -3, 5) and (7, 8, 7).

13. Find the vector projection of the vector $7\hat{i} + \hat{j} - 4\hat{k}$ on (i) $2\hat{i} + 6\hat{j} + 3\hat{k}$
(ii) $7\hat{i} + \hat{j} - 3\hat{k}$ (CBSE 2016)

14. Let $\vec{u}, \vec{v}$ and $\vec{w}$ be vectors such that $\vec{u} + \vec{v} + \vec{w} = \vec{0}$. If $|\vec{u}| = 3, |\vec{v}| = 4$ and $|\vec{w}| = 5$, then find $\vec{u}, \vec{v} + \vec{v}, \vec{w} + \vec{w}$, $\vec{u}$. (IIT 1995)

15. Find the angles which the vector $\vec{a} = \hat{i} - \hat{j} + \sqrt{2}\hat{k}$ makes with the coordinate axes.

16. Dot product of a vector with $\hat{i}+\hat{j}-3\hat{k}, \hat{i}+3\hat{j}-2\hat{k}$ and $2\hat{i} + \hat{j} + 4\hat{k}$ are 0,5 and 8 respectively. Find the vector.
(CBSE 2003)

17. If the sum of two unit vectors is a unit vector, prove that the magnitude of their difference is $\sqrt{3}$.

18. If $\hat{a}$ and $\hat{b}$ are unit vectors inclined at an angle θ, then prove that
(i) $\cos\frac{\theta}{2} = \frac{1}{2}|\hat{a} + \hat{b}|$,
(ii) $\tan\frac{\theta}{2} = \frac{|\hat{a}-\hat{b}|}{|\hat{a}+\hat{b}|}$.

19. If $\vec{a}, \vec{b}, \vec{c}$ are three murually perpendicular unit vectors, then prove that $|\vec{a}+\vec{b}+\vec{c}| = \sqrt{3}$.
(Type ISC 2020)

20. If $|\vec{a}+\vec{b}| = 60, |\vec{a}-\vec{b}| = 40$ and $|\vec{b}| = 46$, find $|\vec{a}|$.

21. If a vector $\vec{a}$ is perpendicular to two non-collinear vectors $\vec{b}$ and $\vec{c}$, then $\vec{a}$ is perpendicular to every vector in the plane of $\vec{b}$ and $\vec{c}$.

22. If $\vec{A} = \hat{i}+2\hat{j}+3\hat{k}, \vec{B} = -\hat{i}+2\hat{j}+\hat{k}$ and $\vec{C} = 3\hat{i}+\hat{j}$, find t such that $\vec{A}+t\vec{B}$ is perpendicular to $\vec{C}$.

23. If A, B, C, D are any four points, show that $\overrightarrow{AB}\cdot\overrightarrow{CD}+\overrightarrow{BC}\cdot\overrightarrow{AD}+\overrightarrow{CA}\cdot\overrightarrow{BD} = 0$

24. If $\vec{a}$ and $\vec{b}$ are vectors with a, b as their respective lengths, show that $\left(\frac{\vec{a}}{a^2}-\frac{\vec{b}}{b^2}\right)^2 = \left(\frac{\vec{a}-\vec{b}}{ab}\right)^2$.

Section B: Vector (Cross) Product — Solved Examples (Questions)

Example 29. If $\overrightarrow{a} = -\hat{i}+3\hat{j}+\hat{k}$ and $\overrightarrow{b} = \hat{i}+2\hat{j}+5\hat{k}$, prove that $\overrightarrow{a}\times\overrightarrow{b}$ represents a vector which is perpendicular to both $\overrightarrow{a}$ and $\overrightarrow{b}$. (Type ISC 2019)

Example 30. Find the sine of the angle between the vectors $\hat{i}+3\hat{j}+2\hat{k}$ and $2\hat{i}-4\hat{j}+\hat{k}$. (ISC 2007)

Example 31. (i) Find $|\vec{a}\times\vec{b}|$ if $|\vec{a}| = 10, |\vec{b}| = 2, \vec{a}\cdot\vec{b} = 12$.
(ii) Find $\vec{a}\cdot\vec{b}$ if $|\vec{a}| = 2, |\vec{b}| = 5$ and $|\vec{a}\times\vec{b}| = 8$. (ISC 2008)

Example 32. Let $\vec{a}, \vec{b}, \vec{c}$ be unit vectors such that $\vec{a}\cdot\vec{b} = \vec{a}\cdot\vec{c} = 0$ and the angle between $\vec{b}$ and $\vec{c}$ is $\frac{\pi}{6}$, prove that $\vec{a} = \pm 2(\vec{b}\times\vec{c})$.

Example 33. Find a unit vector perpendicular to the plane of two vectors $\overrightarrow{a}$ and $\overrightarrow{b}$, where $\vec{a} = \hat{i}-\hat{j}+2\hat{k}$ and $\vec{b} = 2\hat{i}+3\hat{j}-\hat{k}$. (ISC Type 2016, 2007, 2004, 1998)

Example 34. If $\overrightarrow{a} = 2\hat{i}+\hat{j}-3\hat{k}$ and $\overrightarrow{b} = \hat{i}-2\hat{j}+\hat{k}$, find a vector of magnitude 5 perpendicular to both $\overrightarrow{a}$ and $\overrightarrow{b}$.
(NDA April 2006 Type)

Example 35. Find a unit vector perpendicular to the plane ABC where A, B and C are the points (3, -1, 2), (1, -1, -3) and (4, -3, 1) respectively.

Example 36. What inference can you draw, if $\overrightarrow{a}\times\overrightarrow{b} = \overrightarrow{0}$ and $\overrightarrow{a}\cdot\overrightarrow{b} = 0$.

Example 37. Prove that for any two vectors $\overrightarrow{a}$ and $\overrightarrow{b}$, $\overrightarrow{a}\cdot(\overrightarrow{a}\times\overrightarrow{b}) = 0$.

Example 38. If $\overrightarrow{b} = \overrightarrow{c}$, should $\overrightarrow{a}\times\overrightarrow{b} = \overrightarrow{c}\times\overrightarrow{a}$?

Example 39. If $\vec{a}\times\vec{b} = \vec{c}\times\vec{d}, \vec{a}\times\vec{c} = \vec{b}\times\vec{d}$, show that $\vec{a}-\vec{d}$ is parallel to $\vec{b}-\vec{c}$.
(CBSE 2009, 01)

Example 40. Find the area of the parallelogram whose adjacent sides are determined by the vectors $\vec{a} = \hat{i}+2\hat{j}+3\hat{k}$ and $\vec{b} = -3\hat{i}-2\hat{j}+\hat{k}$. (Type ISC Sample Paper 2018, Roorkee)

Example 41. Show that the area of a parallelogram having diagonals $3\hat{i}+\hat{j}-2\hat{k}$ and $\hat{i}-3\hat{j}+4\hat{k}$ is $5\sqrt{3}$.
(ISC 2006 Type, 1999)

Example 43. Find the area of a triangle whose two adjacent sides are given by $a = \hat{i}+4\hat{j}-\hat{k}$ and $\vec{b} = \hat{i}+\hat{j}+2\hat{k}$.

Example 44. Find the area of the $\triangle ABC$, whose vertices are $A(1,-1,2), B(2,1,-1)$ and $C(3,-1,2)$.
(IIT)

Example 45. Find the area of the triangle formed by the points $A(1,2,3), B(2,3,1)$ and $C(3,1,2)$.

Example 46. If D, E, F are the mid points of the sides BC, CA and AB respectively, then prove by vector method that $\triangle DEF = \frac{1}{4}(\triangle ABC)$. (ISC 2001)

Example 47. Prove that the points whose position vectors are $\overrightarrow{a}$, $\overrightarrow{b}$ and $3\overrightarrow{a}-2\overrightarrow{b}$ are collinear.

Example 49. and $\overrightarrow{b}$ are two vectors satisfying the condition $\overrightarrow{a}\times\overrightarrow{b} = \overrightarrow{0}$. Does it imply that one of the vectors $\overrightarrow{a}$ or $\overrightarrow{b}$ must be a null vector ? Give one example in support of your answer.

Example 50. Does $\overrightarrow{a}\times\overrightarrow{b} = \overrightarrow{a}\times\overrightarrow{c}$ always mean $\overrightarrow{b} = \overrightarrow{c}$ in vector algebra? Give reasons with your answer. (Roorkee)

Example 51. If $\overrightarrow{a}+\overrightarrow{b}+\overrightarrow{c} = \overrightarrow{0}$ then prove that $\overrightarrow{a}\times\overrightarrow{b} = \overrightarrow{b}\times\overrightarrow{c} = \overrightarrow{c}\times\overrightarrow{a}$.
(AIEEE 2002 Type, Roorkee)

Example 52. Prove that $\vec{a}\times(\vec{b}+\vec{c})+\vec{b}\times(\vec{c}+\vec{a})+\vec{c}\times(\vec{a}+\vec{b}) = \overrightarrow{0}$. (NMOC, Roorkee)

Example 53. Show that $(\vec{a}-\vec{b})\times(\vec{a}+\vec{b}) = 2(\vec{a}\times\vec{b})$ and give a geometrical interpretation to $|(\vec{a}-\vec{b})\times(\vec{a}+\vec{b})| = 2|(\vec{a}\times\vec{b})|$.
Hence, find the area of a parallelogram whose diagonals are vectors $\overrightarrow{a} = 3\hat{i}+\hat{j}-2\hat{k}$ and $\vec{b} = \hat{i}-3\hat{j}+4\hat{k}$. (ISC 1999, NMOC, Dhanbad 1995)

Example 54. Find the perpendicular distance of $A(1,4,-2)$ from BC where coordinates of B and C are respectively (2, 1, -2) and (0, - 5, 1).

Example 55. If A, B, C are three non-collinear points with position vectors $\overrightarrow{a}, \overrightarrow{b}, \overrightarrow{c}$ respectively, then show that the length of the perpendicular from C on AB is

$$\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}$$

(ISC 2018)

Example 56. $\overrightarrow{a}, \overrightarrow{b}, \overrightarrow{c}$ are three vectors such that $\overrightarrow{a}\times\overrightarrow{b} = \overrightarrow{c}, \overrightarrow{b}\times\overrightarrow{c} = \overrightarrow{a}$. Prove that $\overrightarrow{a}, \overrightarrow{b}, \overrightarrow{c}$ are mutually at right angle and $|\vec{b}| = 1, |\vec{c}| = |\vec{a}|$.

Example 57. If $\vec{A} = (1,1,1), \vec{C} = (0,1,-1)$ are given vectors, then find a vector $\vec{B}$ satisfying the equation $\vec{A}\times\vec{B} = \vec{C}$ and $\vec{A}\cdot\vec{B} = 3$. (Type CBSE 2008, IIT)

EXERCISE 22 (b)

1. Find $\overrightarrow{a} \times \overrightarrow{b}$, when
(i) $\vec{a} = 3\hat{i} - \hat{j} + \hat{k}, \vec{b} = 2\hat{i} + \hat{j} - \hat{k}$
(ii) $\vec{a} = 2\hat{i} - 3\hat{j} - \hat{k}, \vec{b} = \hat{i} + 4\hat{j} - 2\hat{k}$
(iii) In (ii), verify that $\vec{a} \times \vec{b} \neq \vec{b} \times \vec{a}$
(iv) $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}, \vec{b} = 3\hat{i} + 4\hat{j} - \hat{k}$, verify that $\vec{a} \times \vec{b}$ is perpendicular to both $\vec{a}$ and $\vec{b}$
(v) Compute $\vec{a} \times (\vec{b} + \vec{c})$, if $\vec{a} = 7\hat{i} - 2\hat{j} + 3\hat{k}, \vec{b} = 2\hat{i} + 8\hat{k}$, $\vec{c} = \hat{i} + \hat{j} + \hat{k}$

2. Find $|\vec{a} \times \vec{b}|$, when
(i) $\vec{a} = \hat{i} + 3\hat{j} - 2\hat{k}, \vec{b} = -\hat{i} + 3\hat{k}$
(ii) $\vec{a} = 2\hat{i} + \hat{k}, \vec{b} = \hat{i} + \hat{j} + \hat{k}$

3. If $\vec{a} = 2\hat{i} + 3\hat{j}, \vec{b} = -\hat{i} + 3\hat{j} + \hat{k}$ and $\vec{c} = \hat{i} + 2\hat{j} + 5\hat{k}$ be three vectors, find
(i) $\vec{a} \times \vec{b}$
(ii) $\vec{b} \times \vec{c}$
(iii) $(\vec{a} - \vec{b}) \times (\vec{c} - \vec{b})$

4. If $|\vec{a}| = \sqrt{26}, |\vec{b}| = 7$ and $|\vec{a} \times \vec{b}| = 35$, find $\vec{a} \cdot \vec{b}$.

5. If $\vec{a} = 3\hat{i} - \hat{j} - 2\hat{k}$ and $\vec{b} = 2\hat{i} + 3\hat{j} + \hat{k}$, find $(\vec{a} + 2\vec{b}) \times (2\vec{a} - \vec{b})$.

6. Find a unit vector perpendicular to the plane of the vectors.
(i) $-3\hat{j} + 4\hat{k}$ and $4\hat{i} + 3\hat{j}$
(ii) $\vec{a} = 2\hat{i} + \hat{j} + \hat{k}$ and $\vec{b} = \hat{i} + 2\hat{j} + \hat{k}$.

7. (i) Find the vector of magnitude 9, which is perpendicular to both the vectors $4\hat{i} - \hat{j} + 3\hat{k}$ and $-2\hat{i} + \hat{j} - 2\hat{k}$.
(ii) Find a vector whose length is 3 and which is perpendicular to both the vectors $\vec{a} = 3\hat{i} + \hat{j} - 4\hat{k}$ and $\vec{b} = 6\hat{i} + 4\hat{j} - 2\hat{k}$.

8. (i) If the position vectors of the three points A, B, C are respectively $\hat{i} + \hat{j} + \hat{k}, 2\hat{i} + 3\hat{j} - 4\hat{k}$ and $7\hat{i} + 4\hat{j} + 9\hat{k}$, find the unit vector perpendicular to the plane of the triangle ABC.
(ii) Find the unit vectors perpendicular to the plane ABC, when the position vectors of A, B and C are $2\hat{i} - \hat{j} + \hat{k}, \hat{i} + \hat{j} + 2\hat{k}$ and $2\hat{i} + 3\hat{k}$ respectively.
(iii) Find a unit vector perpendicular to the plane determined by the points, $P(1, -1, 2)$, $Q(2, 0, -1)$ and $R(0, 2, 1)$. (III)

9. Find the sine of the angle between the vectors.
(i) $\vec{a} = 3\hat{i} - 4\hat{j} + 5\hat{k}, \vec{b} = \hat{i} - \hat{j} + \hat{k}$
(ii) $\hat{i} + \hat{j}$ and $\hat{j} + \hat{k}$

10. Find the angle between two vectors $\vec{a}$ and $\vec{b}$, if $|\vec{a} \times \vec{b}| = \vec{a} \cdot \vec{b}$.

11. If $|\overrightarrow{a}| = 2, |\overrightarrow{b}| = 7$ and $\overrightarrow{a} \times \overrightarrow{b} = 3\hat{i} + 2\hat{j} + 6\hat{k}$, find the angle between $\overrightarrow{a}$ and $\overrightarrow{b}$.

12. If θ is the angle between $\vec{a}$ and $\vec{b}$. Then prove that $\tan\theta = \frac{|\vec{a} \times \vec{b}|}{\vec{a} \cdot \vec{b}}$.

13. If $\vec{a} = 4\hat{i} - \hat{j} + \hat{k}, \vec{b} = \hat{i} + \hat{j} + \hat{k}$ and $\vec{c} = \hat{i} - \hat{j} + \hat{k}$, verifythat, $\vec{a} \times (\vec{b} + \vec{c}) = (\vec{a} \times \vec{b} + \vec{a} \times \vec{c})$.

14. Find the area of the parallelogram whose adjacent sides are
(i) $\hat{i} - 3\hat{j} + \hat{k}$ and $\hat{i} + \hat{j} + \hat{k}$
(CBSE 2002 C)
(ii) $\hat{i} + 2\hat{j} + 3\hat{k}$ and $3\hat{i} - 2\hat{j} + \hat{k}$
(iii) $2\hat{i} + \hat{j} + 3\hat{k}$ and $\hat{i} - \hat{j}$
(HB 1991)
(iv) $2\hat{i}$ and $3\hat{j}$

15. Find the area of a parallelogram whose diagonals are $3\hat{i} + 4\hat{j}$ and $\hat{i} + \hat{j} + \hat{k}$.

16. If $\vec{a} = 2\hat{i} - 3\hat{j} + \hat{k}, \vec{b} = -\hat{i} + \hat{k}, \vec{c} = 2\vec{j} - \vec{k}$, find the area of the parallelogram having diagonals $\vec{a} + \vec{b}$ and $\vec{b} + \vec{c}$.

17. If $\vec{p}$ and $\vec{q}$ are unit vectors forming angle of 30°; find the area of the parallelogram having $\vec{a} = \vec{p} + 2\vec{q}$ and $\vec{b} = 2\vec{p} + \vec{q}$ as its diagonals

18. Find the area of the triangle whose adjacent sides are determined by the vectors
(i) $\overrightarrow{OA} = 3\hat{i} + 2\hat{j} - \hat{k}, \overrightarrow{OB} = \hat{i} + 3\hat{j} + \hat{k}$
(ii) $3\hat{i} + 4\hat{j}$ and $-5\hat{i} + 7\hat{j}$

19. Find by vector method, the area of triangle ABC whose vertices are
(i) $A(1, 3, 2), B(2, -1,$
$1)$ and $C(-1, 2, 3)$
(ii) $A(1, 2, 3), B(2, 5, -1)$ and $C(-1, 1, 2)$.
[Type ISC 2020]

20. Show that the points whose position vectors are given below, are collinear
(i) $\vec{a} - 2\vec{b} + 3\vec{c}, 2\vec{a} + 3\vec{b} - 4\vec{c}$ and $-7\vec{b} + 10\vec{c}$
(ii) $5\hat{i} + 6\hat{j} + 7\hat{k}, 7\hat{i} - 8\hat{j} + 9\hat{k}$ and $3\hat{i} + 20\hat{j} + 5\hat{k}$

21. Find λ such that $\vec{a} = \hat{i} + \lambda\hat{j} + 3\hat{k}$ and $\vec{b} = 3\hat{i} + 2\hat{j} + 9\hat{k}$ are parallel.

22. Find the values of a for which the vectors $3\hat{i} + 2\hat{j} + 9\hat{k}$ and $\hat{i} + a\hat{j} + 3\hat{k}$ are
(i) perpendicular
(ii) parallel.

23. Find a unit vector parallel to the ty plane and perpendicular to the vector $4\hat{i} - 3\hat{j} + \hat{k}$.

24. By vector method obtain the perpendicular distance of the point $(6, -4, 4)$ from the line passing through the points $(2, 1, 2)$ and $(3, -1, 4)$.

25. Given, $\vec{a} = \frac{1}{7}(2\hat{i} + 3\hat{j} + 6\hat{k}), \vec{b} = \frac{1}{7}(3\hat{i} - 6\hat{j} + 2\hat{k})$, $\vec{c} = \frac{1}{7}(6\hat{i} + 2\hat{j} - 3\hat{k}), \hat{i}, \hat{j}, \hat{k}$ being a right handed orthogonal system of unit vectors in space, show that $\vec{a}, \vec{b}, \vec{c}$ is also another system of unit vectors mutually perpendicular.

26. $\vec{a}, \vec{b}, \vec{c}$ represent three sides $\overrightarrow{BC}, \overrightarrow{CA}$ and $\overrightarrow{AB}$ of a $\triangle ABC$ such that $|\vec{a}| = 13, |\vec{b}| = 14, |\vec{c}| = 15$. Find $\angle C$.

27. $\vec{a}, \vec{b}, \vec{c}$ are three vectors such that $\vec{a} \cdot \vec{b} = \vec{a} \cdot \vec{c}; \vec{a} \times \vec{b} = \vec{a} \times \vec{c}, \vec{a} \neq 0$ prove that $\vec{b} = \vec{c}$.

28. If $r = x\hat{i} + y\hat{j} + z\hat{k}$, then show that
(i) $(\vec{r} \times i)^2 = y^2 + z^2$ and
(ii) $(\vec{r} \times \hat{i}) \cdot (\vec{r} \times \hat{j}) + xy = 0$.
(CBSE 2015)

29. For any vector $\vec{a}$, prove that

$$\left|\vec{a} \times i\vec{f} + |\vec{a} \times \hat{j}|^2 + |\vec{a} \times \hat{k}|^2 = 2\left|\vec{a}\right|^2\right.$$

(Civil Services 2006)

Section C: Scalar Triple Product — Solved Examples (Questions)

Example 58. Prove that $[\hat{i}\hat{j}\hat{k}] = 1$ and $[\hat{i}\hat{k}\hat{j}] = -1$.

Example 59. If $\vec{a} = 2\hat{i} + \hat{j} + 3\hat{k}, \vec{b} = -\hat{i} + 2\hat{j} + \hat{k}$ and $\vec{c} = 3\hat{i} + \hat{j} + 2\hat{k}$, find $[\vec{a}\vec{b}\vec{c}]$.

Example 60. Evaluate $\hat{\boldsymbol{i}} - (\hat{\boldsymbol{j}} \quad \hat{\boldsymbol{k}}) + (\hat{\boldsymbol{i}} \quad \hat{\boldsymbol{k}}) \cdot \hat{\boldsymbol{j}}$

Example 61. Find the volume of the parallelopiped whose coterminous edges are represented by the vectors $\boldsymbol{2\hat{i} - 3\hat{j}}, \hat{\boldsymbol{i}} + \hat{\boldsymbol{j}} - \hat{\boldsymbol{k}}$ and $\boldsymbol{3\hat{i} - \hat{k}}$.
(Type ISC 2015, IIT, Type 1996, 2001, AMU 2002)

Example 62. Show that the vectors $\vec{a} = -2\hat{i} - 2\hat{j} + 4\hat{k}$, $\vec{b} = -2\hat{i} + 4\hat{j} - 2\hat{k}$ and $\vec{c} = 4\hat{i} - 2\hat{j}$ are coplanar.

Example 63. Find the value of λ which makes the vectors $\hat{\boldsymbol{i}} - \hat{\boldsymbol{j}} + \hat{\boldsymbol{k}}, 2\hat{\boldsymbol{i}} + \hat{\boldsymbol{j}} - \hat{\boldsymbol{k}}$ and $\lambda\hat{\boldsymbol{i}} + \hat{\boldsymbol{j}} + \lambda\hat{\boldsymbol{k}}$ coplanar. (Type CBSE 2015)

Example 64. Show that four points whose position vectors are $\boldsymbol{6\hat{i} - 7\hat{j}}, \boldsymbol{16\hat{i} - 29\hat{j} - 4\hat{k}}, \boldsymbol{3\hat{j} - 6\hat{k}}, \boldsymbol{2\hat{i} + 5\hat{j} + 10\hat{k}}$ are coplanar.

Example 65. Show that the four points $\boldsymbol{A(4, 5, 1)}, \boldsymbol{B(0, -1, -1)}, \boldsymbol{C(3, 9, 4)}$ and $\boldsymbol{D(-4, 4, 4)}$ are coplanar. (ISC 2018)

Example 66. Show that the vectors $\boldsymbol{\vec{a} - 2\vec{b} + 3\vec{c}}$, $\boldsymbol{-2\vec{a} + 3\vec{b} - 4\vec{c}}$ and $\boldsymbol{\vec{a} - 3\vec{b} + 5\vec{c}}$ are coplanar.

Example 67. Find the value of $\boldsymbol{\lambda}$ for which the four points with position vectors $3\hat{i} - 2\hat{j} - \hat{k}, 2\hat{i} + 3\hat{j} - 4\hat{k}$, $-\hat{i} + \hat{j} + 2\hat{k}$ and $4\hat{i} + 5\hat{j} + \lambda\hat{k}$ are coplanar.
(Type ISC 2017, 2010, IIT)

Example 68. (i) If $\vec{a}, \vec{b}, \vec{c}$ be three vectors, prove that $[\vec{a} + \vec{b}, \vec{b} + \vec{c}, \vec{c} + \vec{a}] = 2[\vec{a}, \vec{b}, \vec{c}]$
(ISC 2011, 2007)
Hence, prove that the vectors $\boldsymbol{\vec{a} + \vec{b}, \vec{b} + \vec{c}, \vec{c} + \vec{a}}$ are coplanar iff $\boldsymbol{\vec{a}, \vec{b}, \vec{c}}$ are coplanar.
(ii) Show that $\vec{a}, \vec{b}, \vec{c}$ are non-coplanar if and only if $\vec{a} + \vec{b}$, $\vec{b} + \vec{c}, \vec{c} + \vec{a}$ are noncoplanar.

Example 69. (i) Simplify $[\boldsymbol{\vec{a} - \vec{b}, \vec{b} - \vec{c}, \vec{c} - \vec{a}}]$ or Prove that $[\boldsymbol{\vec{a} - \vec{b}, \vec{b} - \vec{c}, \vec{c} - \vec{a}}] = \boldsymbol{0}$.
(ISC 2012)
(ii) For any three vectors $\vec{a}, \vec{b}, \vec{c}$, show that $\vec{a} - \vec{b}, \vec{b} - \vec{c}$, $\vec{c} - \vec{a}$ are coplanar.
(ISC 2016)

Example 70. Let $\vec{a}, \vec{b}, \vec{c}$ be three non-zero vectors. If $\vec{a} \cdot (\vec{b} \quad \vec{c}) = 0$ and $\vec{b}$ and $\vec{c}$ are not parallel, then prove that $\boldsymbol{\vec{a} = \lambda\vec{b} + \mu\vec{c}}$, where λ and μ are some scalars.

Example 71. If $\vec{a}, \vec{b}, \vec{c}$ are the position vectors of points A, B, C prove that $(\boldsymbol{\vec{a}\vec{b} + \vec{b}\vec{c} + \vec{c}\vec{a}})$ is a vector perpendicular to the plane of triangle $\boldsymbol{ABC}$.

Example 72. Prove that $(\vec{p} - \vec{q}) \cdot [(\vec{q} - \vec{r}) \quad (\vec{r} - \vec{p})] = \boldsymbol{0}$.

Example 73. If $\boldsymbol{\vec{a}}$, $\boldsymbol{\vec{b}}$ and $\boldsymbol{\vec{c}}$ are perpendicular to each other, then prove that

$$(\vec{a} \cdot \vec{b} \quad \vec{c})^2 = a^2b^2c^2.$$

Example 74. Prove that $\boldsymbol{\vec{a} \cdot (\vec{b} + \vec{c}) \quad (\vec{a} + \vec{b} + \vec{c}) = 0}$.

Example 75. Prove that $\boldsymbol{\vec{a} \cdot [(\vec{b} + \vec{c}) \quad (\vec{a} + 2\vec{b} + 3\vec{c})] = [\vec{a}\,\vec{b}\,\vec{c}]}$.
(Type ISC 2020, ISC 2014, 2005, Sample Paper 2018)

EXERCISE 22 (c)

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1. Determine $[\vec{a}, \vec{b}, \vec{c}]$, if $\vec{a} = 2\hat{i} - 3\hat{j}, \vec{b} = \hat{i} + \hat{j} - \hat{k}, \vec{c} = 3\hat{i} - \hat{k}$

2. If $\hat{i}, \hat{j}, \hat{k}$ is an orthogonal unit vector triad in a right handed system, then prove that
(i) $[\hat{i}\hat{j}\hat{k}] = [\hat{j}\hat{k}\hat{i}] = [\hat{k}\hat{i}\hat{j}] = 1$
(ii) $[\hat{i}\hat{k}\hat{j}] = [\hat{j}\hat{i}\hat{k}] = [\hat{k}\hat{j}\hat{i}] = -1$
(iii) $[\hat{i}\hat{j}\hat{k}] + [\hat{i}\hat{k}\hat{j}] = 0$.
(iv) $[\hat{i}\hat{k}\hat{j}] + [\hat{k}\hat{j}\hat{i}] + [\hat{j}\hat{k}\hat{i}] = -1$ (UPSEAT 2002)

3. If α and β are any vectors. Prove that $\vec{\beta} \cdot (\vec{\alpha} \times \vec{\beta}) = \vec{0}$.

4. Find the volume of the parallelopiped whose edges are represented by the vectors
(i) $\vec{a} = 2\hat{i} - 3\hat{j} + \hat{k}, \vec{b} = \hat{i} - \hat{j} + 2\hat{k}$ and $\vec{c} = 2\hat{i} + \hat{j} - \hat{k}$
(ii) $\vec{a} = 11\hat{i}, \vec{b} = 2\hat{j}, \vec{c} = 13\hat{k}$

5. The volume of a parallelopiped whose edges are represented by $-12\hat{i} + \lambda\hat{k}, 3\hat{j} - \hat{k}$, $2\hat{i} + \hat{j} - 15\hat{k}$ is 546. Find the value of λ. (MNR)

6. $A(1, 1, 1), B(2, 1, 3), C(3, 2, 2)$ and $D(3, 3, 4)$. Find the volume of the parallelopiped of which the segments AB, AC and AD are coterminous edges.

7. Prove that $[i - j, j - k, k - i] = 0$.

8. Show that the following vectors are coplanar:

$$\vec{a} = \hat{i} + 2\hat{j} - \hat{k}, \vec{b} = 3\hat{i} + 2\hat{j} + 7\hat{k}, \vec{c} = 5\hat{i} + 6\hat{j} + 5\hat{k}$$

9. If $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}, \vec{b} = \hat{i} - 3\hat{j} - 5\hat{k}, \vec{c} = 3\hat{i} - 4\hat{j} - 4\hat{k}$, determine $[\vec{a}, \vec{b}, \vec{c}]$ and interpret the result.

10. Find the value of λ such that the following vectors are coplanar:
(i) $2\hat{i} - 4\hat{j} + 5\hat{k}, \hat{i} - 2\hat{j} + \hat{k}, 3\hat{i} + 2\hat{j} - 5\hat{k}$
(ii) $\vec{a} = \hat{i} - \hat{j} + \hat{k}, \vec{b} = 2\hat{i} + \hat{j} - \hat{k}, \vec{c} = \lambda\hat{i} - \hat{j} + \lambda\hat{k}$

11. Show that the four points having position vectors $\hat{i} - \hat{j} + 2\hat{k}, 6\hat{i} + 11\hat{j} + 2\hat{k}, \hat{i} + 2\hat{j} + 6\hat{k}$, $\hat{i} + \frac{1}{2}\hat{j} + 4\hat{k}$ are coplanar.

12. (i) Show that the points $A(-1,4,-3), B(3,2,-5), C(-3,8,-5)$ and $D(-3,2,1)$ are coplanar.
(ii) Show that the four points $-\vec{a}+4\vec{b}-3\vec{c}, 3\vec{a}+2\vec{b}-5\vec{c}, -3\vec{a}+8\vec{b}-5\vec{c}$ and $-3\vec{a}+2\vec{b}+\vec{c}$ are coplanar.

13. Find the value of λ so that the four points with position vectors $-\hat{j}+\hat{k}, 2\hat{i}-\hat{j}-\hat{k}, \hat{i}+\lambda\hat{j}+\hat{k}$ and $3j+3k$ are coplanar. (Type ISC 2010)

14. Show that $\vec{a},\vec{b},\vec{c}$ are coplanar if and only if $\vec{b}+\vec{c}$, $\vec{c}+\vec{a}, \vec{a}+\vec{b}$ are coplanar.
(CBSE Delhi 2016)

15. Simplify: (i) $\vec{a}\cdot(\vec{a}\times\vec{b})$ (ii) $(\vec{b}+\vec{c})\cdot[(\vec{c}+\vec{a})\times(\vec{a}+\vec{b})]$

16. Find a unit vector coplanar with $\hat{i}+\hat{j}+2\hat{k}$ and $\hat{i}+2\hat{j}+\hat{k}$. perpendicular to $\hat{i}+\hat{j}+\hat{k}$.
(IIT 1992)

17. If $\vec{A},\vec{B},\vec{C}$ are three non-coplanar vectors, then prove that $\frac{\vec{A}\cdot(\vec{B}\times\vec{C})}{(\vec{C}\times\vec{A})\cdot\vec{B}}+\frac{\vec{B}\cdot(\vec{A}\times\vec{C})}{\vec{C}\cdot(\vec{A}\times\vec{B})}=0$.
(III)

FOR SELF ASSESSMENT AND REVISION (ISC QUESTIONS)

1. The diagonals AC and BD of a parallelogram $ABCD$ are represented by the vectors $\hat{i}+\hat{j}-\hat{k}$ and $\hat{i}-\hat{j}+\hat{k}$ respectively. Find the area of the parallelogram. (ISC 1993)

2. Determine the value of λ so that the vectors $2\hat{i}-3\hat{j}+\hat{k}$, $\hat{i}+2\hat{j}-3\hat{k}$ and $\hat{j}+\lambda\hat{k}$ are coplanar. (ISC 1994)

3. The position vectors of the vertices A, B and C of a triangle ABC are $\vec{a},\vec{b}$ and $\vec{c}$. Show that the vector area of the triangle is $\frac{1}{2}[\vec{b}\times\vec{c}+\vec{c}\times\vec{a}+\vec{a}\times\vec{b}]$ (ISC 1994)

4. Given, $\vec{a}=3\hat{i}-\hat{j}, \vec{b}=2\hat{i}+\hat{j}-3\hat{k}$, express $\vec{b}$ as $\vec{b}_1+\vec{b}_2$, where $\vec{b}_1$ is parallel to $\vec{a}$ and $\vec{b}_2$ is perpendicular to $\vec{a}$.
(ISC 1995)

5. Find the volume of the parallelopiped whose edges are represented by $\vec{a}=2\hat{i}-3\hat{j}+4\hat{k}$, $\vec{b}=\hat{i}+2\hat{j}-\hat{k}$ and $\vec{c}=3\hat{i}-\hat{j}+2\hat{k}$. (ISC 2001, 1996)

6. Find the unit vector perpendicular to the plane of vectors $\vec{a}$ and $\vec{b}$, where $\vec{a}=\hat{i}-\hat{j}+\hat{k}$ and $\vec{b}=\hat{i}+2\hat{j}$. (ISC 1996)

7. Three vectors $\vec{a},\vec{b}$ and $\vec{c}$ are given which are mutually perpendicular and $|\vec{a}|=|\vec{b}|=|\vec{c}|=1$. Show that the vector $\vec{a}+\vec{b}+\vec{c}$ is equally inclined to $\vec{a},\vec{b}$ and $\vec{c}$.
(ISC 1997)

8. Write a unit vector to the plane of two vectors $\vec{x}$ and $\vec{y}$, where $\vec{x}=5\hat{i}+2\hat{j}-3\hat{k}$ and $\vec{y}=-\hat{i}-3\hat{j}-\hat{k}$. (ISC 1998)

9. A particle acted upon by constant forces $4\hat{i}-\hat{j}-3\hat{k}$ and $3\hat{i}+\hat{j}-\hat{k}$ is displaced from the point $\hat{i}+2\hat{j}-3\hat{k}$ to the point $5\hat{i}-4\hat{j}-\hat{k}$. Find the total work done by the forces.
(ISC 1998)

10. Find the area of the parallelogram $ABCD$ where diagonal AC and BD are represented by the vectors $3\hat{i}+\hat{j}-2\hat{k}$ and $\hat{i}-3\hat{j}+4\hat{k}$ respectively. (ISC 2009, 1999)

11. The vectors $2\hat{i}-\hat{j}+\hat{k}, \hat{i}-3\hat{j}-5\hat{k}$ and $3\hat{i}-4\hat{j}-4\hat{k}$ are the position vectors of the vertices A, B and C respectively of the triangle ABC. Prove that ABC is a right-angled triangle.
(ISC 1999)

12. If $\vec{a},\vec{b},\vec{c},\vec{d}$ are the position vectors of four points A, B, C, D respectively, and if $\vec{b}-\vec{a}=\vec{c}-\vec{d}$, show that $ABCD$ is a parallelogram. (ISC 2000)

13. If D, E, F are the mid-points of the sides BC, CA, AB respectively of a triangle ABC, show that the area of $\triangle DEF=\frac{1}{4}$ (area of $\triangle ABC$). (ISC 2011, 2001)

14. The vectors $\vec{a}=3\hat{i}+x\hat{j}-\hat{k}$ and $\vec{b}=2\hat{i}+\hat{j}+y\hat{k}$ are mutually perpendicular. Given that $|\vec{a}|=|\vec{b}|$, find the values of x and y. (ISC 2002)

15. The position vectors of the vertices of a triangle are given as $2\hat{i}-\hat{j}+\hat{k}, \hat{i}-3\hat{j}-5\hat{k}$ and $3\hat{i}-4\hat{j}-4\hat{k}$. Prove that it is a right-angled triangle. (ISC 2003)

16. Find the volume of the parallelopiped whose coterminous edges are represented by the vectors $\hat{i}+\hat{j}+\hat{k}$, $\hat{i}-\hat{j}+\hat{k}$, and $\hat{i}+3\hat{j}-\hat{k}$,. (ISC 2003)

17. If $\vec{a}$ and $\vec{b}$ are two vectors such that $|\vec{a}+\vec{b}|=|\vec{a}|$, prove that $(2\vec{a}+\vec{b})$ is perpendicular to $\vec{b}$. (ISC 2004)

18. Find the unit vector perpendicular to the two vectors $\hat{i}+2\hat{j}-\hat{k}$ and $2\hat{i}+3\hat{j}+\hat{k}$.
(ISC 2004)

19. If $\vec{a}$ and $\vec{b}$ are unit vectors such that $2\vec{a}-4\vec{b}$ and $10\vec{a}+8\vec{b}$ are perpendicular to each other, find the angle between the vectors $\vec{a}$ and $\vec{b}$. (ISC 2005)

20. Prove that $\vec{a}\cdot(\vec{b}+\vec{c})\times(\vec{a}+2\vec{b}+3\vec{c})=[\vec{a}\vec{b}\vec{c}]$. (ISC 2005)

21. The vectors $-2\hat{i}+4\hat{j}+4\hat{k}$ and $-4\hat{i}-2\hat{k}$ represent the diagonals BD and AC of a parallelogram $ABCD$. Find the area of the parallelogram. (ISC 2006)

22. The vectors $\hat{i}+3\hat{j}, 5\hat{k}$ and $\lambda\hat{i}-\hat{j}$ are coplanar. Find the value of λ. (ISC 2006)

23. If $\vec{a},\vec{b},\vec{c}$ are three vectors, show that $(\vec{a}+\vec{b}),\{(\vec{b}+\vec{c})\times(\vec{c}+\vec{a})\}=2[\vec{a}\vec{b}\vec{c}]$.
(ISC 2011, 2007)

24. Find a unit vector perpendicular to the vector $4\hat{i}+3\hat{j}+\hat{k}$ and $2\hat{i}-\hat{j}+2\hat{k}$. Determine the sine of the angle between these two vectors. (ISC 2007)

25. Find $\vec{a}\cdot\vec{b}$, if $|\vec{a}|=2, |\vec{b}|=5$ and $|\vec{a}\times\vec{b}|=8$. (ISC 2008)

26. Find the value of λ for which the four points with position vectors $2\hat{i}+5\hat{j}+\hat{k}, -\hat{j}-4\hat{k}$, $3\hat{i}+\lambda\hat{j}+8\hat{k}$ and $-4\hat{i}+3\hat{j}+4\hat{k}$ are coplanar. (ISC 2010)

27. For any three vectors $\vec{a}, \vec{b}, \vec{c}$ prove:

$$\begin{bmatrix} \vec{a}-\vec{b} & \vec{b}-\vec{c} & \vec{c}-\vec{a} \end{bmatrix} = 0$$

(ISC 2012)

28. If $\vec{a}$ and $\vec{b}$ are unit vectors and θ is the angle between them, then show that $|\vec{a}-\vec{b}| = 2\sin\frac{\theta}{2}$.
(ISC 2013)

29. Find the value of λ for which the four points A, B, C, D will position vectors $-\hat{j}-\hat{k}, 4\hat{i}+5\hat{j}+\lambda\hat{k}, 3\hat{i}+9\hat{j}+4\hat{k}$ and $-4\hat{i}+4\hat{j}+4\hat{k}$ are coplanar. (ISC 2013)

30. Prove that $\vec{a}\cdot[(\vec{b}+\vec{c})\times(\vec{a}+2\vec{b}+3\vec{c})] = [\vec{a}\vec{b}\vec{c}]$. (Type ISC 2020, ISC 2014)

31. Find the volume of a parallelopiped whose edges are represented by the vectors.

$\vec{a} = 2\hat{i}-3\hat{j}-4\hat{k}, \vec{b} = \hat{i}+2\hat{j}-\hat{k}$, and $\vec{c} = 3\hat{i}+\hat{j}+2\hat{k}$

(ISC 2015)

32. For any three vectors $\vec{a}, \vec{b}, \vec{c}$, show that $\vec{a}-\vec{b}, \vec{b}-\vec{c}$, $\vec{c}-\vec{a}$ are coplanar.
(ISC 2016)

33. Find a unit vector perpendicular to each of the vectors $\vec{a}+\vec{b}$ and $\vec{a}-\vec{b}$ where $\vec{a} = 3\hat{i}+2\hat{j}+2\hat{k}, \vec{b} = \hat{i}+2\hat{j}-2\hat{k}$.
(ISC 2016)

VERY SHORT ANSWER AND OBJECTIVE TYPE QUESTIONS

Multiple Choice Questions

1. If α, β, γ are the angles made by a vector with the coordinate axes, then $\sin^2\alpha+\sin^2\beta+\sin^2\gamma =$ ____.
(a) 0
(b) 1
(c) -1
(d) 2

2. The value of λ, for which the vectors $3\hat{i}-6\hat{j}+\hat{k}$ and $2\hat{i}-4\hat{j}+\lambda\hat{k}$ are parallel is
(NCERT Exemplar)
(a) $\frac{2}{3}$
(b) $\frac{3}{2}$
(c) $\frac{5}{2}$
(d) $\frac{2}{5}$

3. The vectors from origin to the points A and B are $\vec{a} = 2\hat{i}-3\hat{j}+2\hat{k}$ and $\vec{b} = 2\hat{i}+3\hat{j}+\hat{k}$ respectively, then area of $\triangle OAB$ is equal to
(NCERT Exemplar)
(a) 340
(b) $\sqrt{25}$
(c) $\sqrt{229}$
(d) $\frac{1}{2}\sqrt{229}$

4. If $|\vec{a}| = 10, |\vec{b}| = 2$ and $\vec{a}\cdot\vec{b} = 12$, then the value of $|\vec{a}\times\vec{b}|$ is
(NCERT Exemplar)
(a) 5
(b) 10
(c) 14
(d) 16

5. A vector of magnitude 7 units, parallel to the resultant of the vectors $\vec{a} = 2\hat{i}-3\hat{j}-2\hat{k}$ and $\vec{b} = -\hat{i}+2\hat{j}+\hat{k}$ is
(a) $\frac{7}{\sqrt{3}}(\hat{i}+\hat{j}+\hat{k})$
(b) $(\hat{i}-\hat{j}-\hat{k})$
(c) $\frac{7}{\sqrt{3}}(\hat{i}-\hat{j}+\hat{k})$
(d) $\frac{7}{\sqrt{3}}(\hat{i}-\hat{j}-\hat{k})$

6. The area (in sq. units) of the parallelogram whose diagonals are along the vectors $8\hat{i}-6\hat{j}$ and $3\hat{i}+4\hat{j}-12\hat{k}$ is
(a) 65
(b) 52
(c) 26
(d) 20

7. If $\vec{a} = 2\hat{i}+3\hat{j}-5\hat{k}, \vec{b} = m\hat{i}+n\hat{j}+12\hat{k}$ and $\vec{a}\times\vec{b} = 0$ then $(m, n) =$
(a) $\left(-\frac{24}{5}, -\frac{36}{5}\right)$
(b) $\left(-\frac{24}{5}, \frac{36}{5}\right)$
(c) $\left(\frac{24}{5}, -\frac{36}{5}\right)$
(d) $\left(\frac{24}{5}, \frac{36}{5}\right)$

8. If $|\vec{a}| = 3, |\vec{b}| = 4$ and the angle between $\vec{a}$ and $\vec{b}$ is $120°$, then $|4\vec{a}+3\vec{b}|$ is equal to
(a) 25
(b) 7
(c) 13
(d) 12

9. If $|\vec{a}| = 2, |\vec{b}| = 5$ and $|\vec{a}\times\vec{b}| = 8$, then $\vec{a}\cdot\vec{b}$ is equal to
(a) 3
(b) 4
(c) 5
(d) 6

10. If the vectors $2\hat{i}+2\hat{j}+6\hat{k}, 2\hat{i}+\lambda\hat{j}+6\hat{k}, 2\hat{i}-3\hat{j}+\hat{k}$ are coplanar, then the value of λ is
(a) -10
(b) 1
(c) 0
(d) 2

11. If $\vec{a}\cdot\vec{b} = \frac{1}{2}|\vec{a}||\vec{b}|$, then the angle between $\vec{a}\cdot\vec{b}$ is
(a) 0°
(b) 30°
(c) 60°
(d) 90°
(CBSE 2020)

12. Let $\vec{a} = \hat{i}-2\hat{j}+3\hat{k}$. If $|\vec{b}|$ is a vector such that $\vec{a}\cdot\vec{b} = |\vec{b}|$ and $|\vec{a}-\vec{b}| = \sqrt{7}$, then $|\vec{b}|$ equals
(a) 7
(b) 14
(c) $\sqrt{7}$
(d) 21
(CBSE 2020)

13. If $(2\hat{i}+6\hat{j}+27\hat{k})\times(\hat{i}-p\hat{j}+q\hat{k})=0$, then the value of p and q are
(a) $p=6, q=27$
(b) $p=3, q=\frac{27}{2}$
(c) $p=6, q=\frac{27}{2}$
(d) $p=3, q=27$
(CBSE Sample Paper 2020)

14. If the angle between $\hat{i}+\hat{k}$ and $\hat{i}+\hat{j}+a\hat{k}$ is $\frac{\pi}{3}$, then the value of a is
(a) 0 or 2
(b) -4 or 0
(c) 0 or -2
(d) 2 or -2

15. If $\vec{u}=\hat{i}+2\hat{j}, \vec{v}=-2\hat{i}+\hat{j}$ and $\vec{w}=4\hat{i}+3\hat{j}$. Find scalars x and y respectively such that $\vec{w}=x\vec{u}+y\vec{v}$.
(a) 4, -2
(b) 2, -1
(c) 3,5
(d) -5, 2

16. The number of vectors of unit lengths perpendicular to the vectors $\vec{a}=\hat{i}+\hat{j}-\hat{k}$ and $\vec{b}=\hat{i}+\hat{j}+\hat{k}$ is
(a) 2
(b) 1
(c) 3
(d) infinite

17. If $2\hat{i}+3\hat{j}, \hat{i}+\hat{j}+\hat{k}$ and $\lambda\hat{i}+4\hat{j}+2\hat{k}$ taken in order are coterminous edges of a parallelepiped of volume 2 cu . units, then value of λ is
(a) -4
(b) 2
(c) 3
(d) 4

Very Short Answer Type Questions

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1. X and Y are two points with position vectors $3\vec{a}+\vec{b}$ and $\vec{a}-3\vec{b}$ respectively. Write the position vector of a point Z which divides the line segment XY in the ratio $2:1$ externally.
(CBSE 2019)

2. Write a unit vector in the direction of the sum of the vectors $\vec{a}=2\hat{i}+2\hat{j}-5\hat{k}$ and $\vec{b}=2\hat{i}+\hat{j}-7\hat{k}$.
(CBSE 2014)

3. Find the area of the parallelogram where adjacent side are the vector $\vec{a}=2\hat{i}+2\hat{j}+3\hat{k}$ and $\vec{b}=-3\hat{i}-2\hat{j}+\hat{k}$.

4. Find $\vec{a}\cdot(\vec{b}\times\vec{c})$, if $\vec{a}=2\hat{i}+\hat{j}+3\hat{k}, \vec{b}=-\hat{i}+2\hat{j}+\hat{k}$ and $\vec{c}=3\hat{i}+\hat{j}+2\hat{k}$.
(AI CBSE 2014)

5. (i) Find the projection of vector $(2\hat{i}-\hat{j}+\hat{k})$ on the vector $(\hat{i}-2\hat{j}+2\hat{k})$.
(CBSE Sample Paper 2014)
(ii) Find the projection of vector $\hat{i}-\hat{j}$ on the vector $\hat{i}+\hat{j}$.
(CBSE 2020)

6. Find the angle between two vectors $\vec{a}$ and $\vec{b}$ with magnitudes S and 3 respectively and when $|\vec{a}\times\vec{b}|=12$. (AI CBSE 2014C)

7. Find the angle between the vectors $\vec{a}=\hat{i}+\hat{j}+\hat{k}$ and $\vec{b}=\hat{i}-\hat{j}+\hat{k}$

8. For what value of λ are the vectors $\vec{a}=2\hat{i}+\lambda\hat{j}+\hat{k}$ and $\vec{b}=\hat{i}-2\hat{j}+3\hat{k}$ perpendicular to each other?

9. If $\vec{a}=x\hat{i}+2\hat{j}-z\hat{k}$ and $\vec{b}=3\hat{i}-y\hat{j}+\hat{k}$ are two equal vectors, then write the value of $x+y+z$.

10. Find the vector of magnitude 5 units in the direction opposite to $(2\hat{i}+3\hat{j}-6\hat{k})$.
(CBSE Sample Paper 2020)

11. If $|\vec{a}\times\vec{b}|^2+|\vec{a}\cdot\vec{b}|^2=144$ and $|\vec{a}|=4$, then find $|\vec{b}|$.
(NCERT Exemplar)

12. Find a vector of magnitude 6 , which is perpendicular to both the vectors $2\hat{i}-\hat{j}+2\hat{k}$ and $4\hat{i}-\hat{j}+3\hat{k}$.
(NCERT Exemplar)

13. Find a unit vector is the direction of $\overrightarrow{PQ}$, where P and Q have coordinates $(5,0,8)$ and $(3,3,2)$ respectively.
(NCERT Exemplar)

14. Find the magnitude of each of the two vectors $\vec{a}$ and $\vec{b}$ having the same magnitude such that the angle between them is $60°$, and their scalar product is $\frac{9}{2}$. (CBSE 2018)